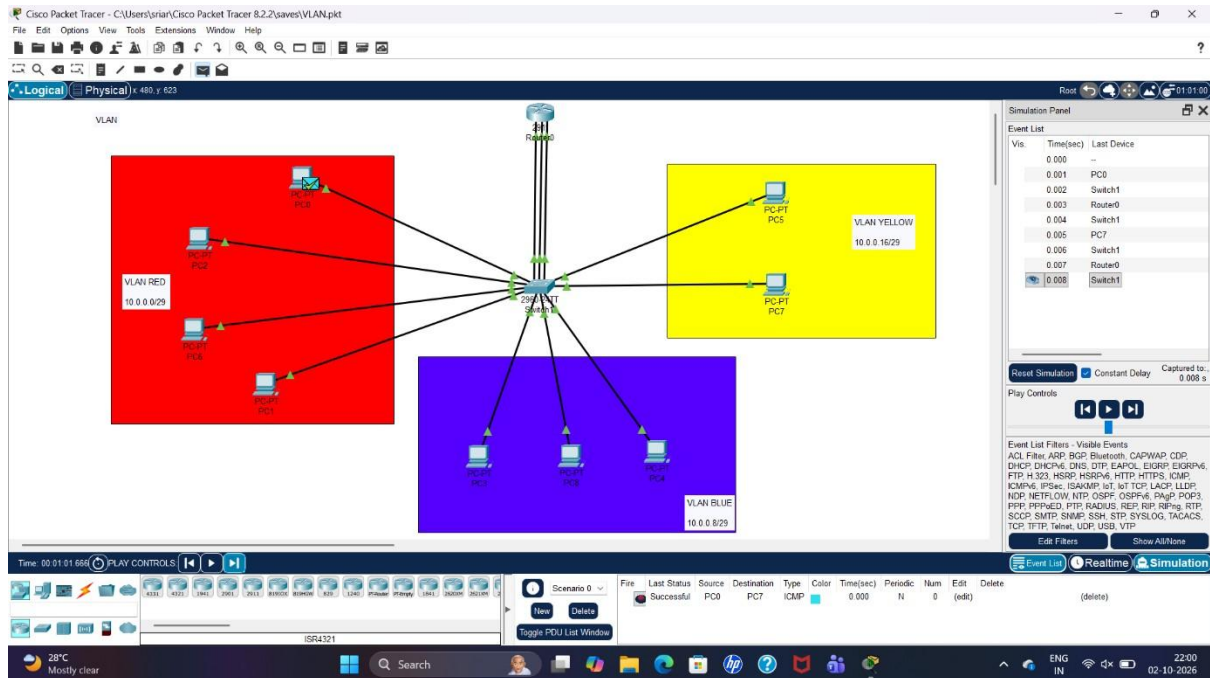


## Project Title: VLAN

Objective: To design and configure a network using three separate VLANs, each associated with a different subnet and connected through a dedicated router link.

### Network Topology:



### Verification:

Device: switch

Command: show vlan brief

```
Switch>show vlan brief
```

| VLAN | Name               | Status | Ports                                                                                                                |
|------|--------------------|--------|----------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/13, Fa0/14, Fa0/15, Fa0/16<br>Fa0/17, Fa0/18, Fa0/19, Fa0/20<br>Fa0/21, Fa0/22, Fa0/23, Fa0/24<br>Gig0/1, Gig0/2 |
| 10   | RED                | active | Fa0/1, Fa0/2, Fa0/3, Fa0/4<br>Fa0/10                                                                                 |
| 11   | blue               | active | Fa0/5, Fa0/6, Fa0/7, Fa0/11                                                                                          |
| 12   | yellow             | active | Fa0/8, Fa0/9, Fa0/12                                                                                                 |
| 1002 | fddi-default       | active |                                                                                                                      |
| 1003 | token-ring-default | active |                                                                                                                      |
| 1004 | fddinet-default    | active |                                                                                                                      |
| 1005 | trnet-default      | active |                                                                                                                      |

```
Switch>
```

Device: Router

Command: Show ip route

```

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/29 is directly connected, GigabitEthernet0/0
L       10.0.0.6/32 is directly connected, GigabitEthernet0/0
C       10.0.0.8/29 is directly connected, GigabitEthernet0/1
L       10.0.0.14/32 is directly connected, GigabitEthernet0/1
C       10.0.0.16/29 is directly connected, GigabitEthernet0/2
L       10.0.0.22/32 is directly connected, GigabitEthernet0/2

```

Ping command from PC in VLAN RED to PC in VLAN YELLOW.

```

PC6
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: FE80::290:CFF:FE77:6A75
    IPv6 Address.....: ::
    IPv4 Address.....: 10.0.0.3
    Subnet Mask.....: 255.255.255.248
    Default Gateway.....: ::
                           10.0.0.6

Bluetooth Connection:

    Connection-specific DNS Suffix.:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                           0.0.0.0

C:\>ping 10.0.0.18

Pinging 10.0.0.18 with 32 bytes of data:

Reply from 10.0.0.18: bytes=32 time=12ms TTL=127
Reply from 10.0.0.18: bytes=32 time=8ms TTL=127
Reply from 10.0.0.18: bytes=32 time=8ms TTL=127
Reply from 10.0.0.18: bytes=32 time=8ms TTL=127

Ping statistics for 10.0.0.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 12ms, Average = 9ms

C:\>

```